

Literature Benchmark Brief: Kruse_2023_PRXEnergy

Title: Physics-informed modeling of power grid frequency dynamics and reserve allocation

Authors: J. Kruse, B. Schael, M. Timme, D. Witthaut (2023)

Venue: PRX Energy | **DOI:** 10.1103/PRXEnergy.2.043003

Cluster: Cluster D: Probabilistic Sizing & Reserve

Core Theoretical Mechanism & Model Formulation

Stochastic swing equation SDEs, primary frequency excursion extremes, reserve sizing

Empirical Scale & Experimental Dataset

European Continental Synchronous Area transmission grid data

Relevance to IEEE TSTE & Sustainable Energy Systems

Links renewable generation forecast errors to grid frequency emergency reserve activation

Identified Gaps & Operational Vulnerabilities

Focuses on bulk transmission grid dynamics rather than wind farm internal telemetry

Role in Manuscript Defense & Reviewer #2 Proof

Delineates Level-1 pre-dispatch screening scope from Level-2 physical grid clearing